

JEE MAIN REVISION MASTER: ENVIRONMENTAL CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 5 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Environmental Chemistry — Core Principles & Governing Laws	4 PYQs	Core Principle
Standard	Environmental Chemistry — Applied Cases, Reactions & Traps	1 PYQs	Applied Topics
Standard	Environmental Chemistry — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Environmental Chemistry — Core Principles & Governing Laws	CFC, CO_2, NO_2

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Environmental Chemistry — Core Principles & Governing Laws (Moderate Yield — 4 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Environmental Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Environmental Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Environmental Chemistry PYQs.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The first chlorinated organic insecticide prepared is..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "Which of the following is not a green house gas?..." [Ans: N/A]
- [2019 Shift] Q#4 (MCQ): "Eutrophication causes..." [Ans: N/A]

3.2 Environmental Chemistry — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Environmental Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "Identify the incorrect statement from the following...." [Ans: N/A]

3.3 Environmental Chemistry — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Environmental Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The first chlorinated organic insecticide prepared is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "Identify the incorrect statement from the following...." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

Identify the incorrect statement from the following.

- (A) Oxides of nitrogen in the atmosphere can cause depletion of the ozone layer.
- (B) Ozone absorbs the intense ultraviolet radiation of Sun.
- (C) Depletion of ozone layer is because of its chemical reactions with chlorofluoro alkanes.
- (D) Ozone absorbs infrared radiation.

Correct Answer: None

Solution Analysis: Option D is incorrect. Ozone primarily absorbs ultraviolet (UV) radiation, particularly in the wavelength range of 240 to 320 nanometers, which helps to protect living organisms from harmful UV rays emitted by the sun. It does not absorb infrared radiation significantly; instead, gases such as carbon dioxide (CO₂), water vapor (H₂O), and methane (CH₄) are primarily responsible for the absorption of infrared radiation, contributing to the greenhouse effect.

Benchmark Problem 2: KCET 2020 (Q#3)

Which of the following is not a green house gas?

- (A) CFC
- (B) CO_2
- (C) O_2
- (D) NO_2

Correct Answer: None

Solution Analysis: Green house gases are the gases which are forming a layer in the atmosphere and prevent the reflected sun rays going upward and as such are responsible for increasing the temperature. CFC, CO_2 and NO_2 are green house gases while O_2 is not a green house gas.